Response-curve family and the biomarker-treatment interaction test in N-of-1 trial simulation: a common-random-numbers comparison of the modified Gompertz template with three alternatives

pmsimstats team

October 28, 2026

Contents

Abstract **2**

1 Introduction **3**

1.1 The question . . . 3
1.2 What the trajectory template is for . . . 3
1.3 How the framework treats the template . . . 5
1.4 Why the question matters . . . 5
1.5 What the N-of-1 simulation literature does . . . 6
1.6 What this paper contributes . . . 6

2 The modified Gompertz function in pmsimstats **7**

3 Alternative response-curve families **7**

3.1 Symmetric logistic . . . 7
3.2 Hyperbolic tangent . . . 8
3.3 Piecewise-linear breakpoint . . . 8
3.4 What the calibration leaves at the sampled visits . . . 8

4 Simulation design **9**

5 Results **10**

5.1 Power across families under common random numbers . . . 10
5.2 Type I error . . . 12
5.3 Bias and sampling distribution of the interaction estimate . . . 13

6 Discussion **13**
6.1 The supported result 13
6.2 Why the covariance architecture is nearly inert 14
6.3 Design conditionality 14
6.4 Conditions that would change the answer 14
6.5 The prazosin reference cell 15
7 Conclusions **15**
Glossary **16**
Abbreviations **16**
8 Conflict of interest **17**
9 Funding **17**
10 Data availability statement **17**
11 Reproducibility **17**
12 References **18**

Abstract

Background. Simulation-based power analyses for aggregated N-of-1 trials need a parametric template for the within-participant response trajectory. The pmsimstats framework uses a modified Gompertz curve for each latent response component. The template is fixed before any analysis is run, and every reported power figure is conditional on it. No published N-of-1 simulation study has compared response-curve families.

Methods. We compared the modified Gompertz curve with three alternatives: the symmetric logistic, the hyperbolic tangent and a piecewise-linear breakpoint model. Each was matched to the Gompertz curve at the asymptote and at a week-5 anchor. We simulated the two-path open-label plus blinded-discontinuation design with $N = 70$ participants, no carryover and the prazosin reference parameters. The interaction was encoded by mean moderation and, separately, by covariance moderation. The four families used common random numbers. We report paired differences in the rejection rate of the biomarker-by-treatment interaction test with paired Monte Carlo standard errors. We report Type I error from 2,000 null replicates per family and bias against the closed-form value $-c_{bm}\sigma_{BR}/\sigma_{bm} = -0.2344$.

Results. At the sampled visits the calibrated logistic curve departs from the Gompertz curve by at most 0.12 points. The two families returned the same decision in all 1,000 seeds under both encodings. The piecewise-linear family differed from Gompertz in 1 seed under mean moderation and 1 under covariance moderation. The hyperbolic tangent is the logistic curve at twice the rate. It had lower power by 0.010 (paired MCSE 0.003) under mean moderation and by 0.008 (paired MCSE 0.003) under covariance moderation. The covariance encoding was nearly but not exactly inert: the families disagreed on the decision in 8 of 1,000 seeds. Type I error was 0.034 (MCSE 0.002), 68 percent of the nominal 0.05. Bias against the closed-form value was within 0.0014 points per biomarker unit in every cell.

Conclusions. Families that coincide at the sampled visits give the same decisions. The only detectable family effect is about one percentage point of power, confined to the hyperbolic tangent, and it does not depend on the interaction encoding. The result is conditional on a visit schedule that samples the discriminating region at two visits. The modified Gompertz default can be retained at this design. Comparisons between data-generating processes should be paired by common random numbers. An earlier unpaired version of this study reported a 0.039 spread that was Monte Carlo noise.

1 Introduction

1.1 The question

The pmsimstats simulation framework supports power analyses for aggregated N-of-1 trials of biomarker-treatment interactions. It operationalizes the design family of Hendrickson et al.[1]. The latent drug-response trajectory in that framework follows a modified Gompertz curve. The Gompertz curve is established in tumor growth dynamics[2,3], microbial growth modeling[4] and the unified-Richards family of growth curves[5–7]. Its three-parameter form represents asymmetric saturation, which linear, exponential and symmetric logistic curves cannot.

This paper asks whether the choice of family changes the inference that the framework reports. We compare the Gompertz curve with three alternatives in a simulation study. The comparators are the symmetric logistic, the hyperbolic tangent and a piecewise-linear breakpoint model. We report power, Type I error and bias of the biomarker-by-treatment interaction estimate under the standard pmsimstats linear mixed-effects analysis.

The authors of the present paper overlap with the authors of the evaluated framework[1]. The present paper evaluates a modeling choice in that framework; it does not re-evaluate the framework's published results.

1.2 What the trajectory template is for

A power simulation for an aggregated N-of-1 trial turns a design into a distribution. The design is a visit schedule with a drug state and an expectancy at each visit. The schedule carries no magnitudes. The trajectory template maps elapsed time onto an expected symptom

reduction. It is the only place in the data-generating process (DGP) where the size and time course of the treatment response are set.

The template is applied once per latent component, and each application reads a different clock. The drug-response component BR is evaluated at cumulative time on drug, t_{od}. The placebo-response component PB is evaluated at accumulated time under nonzero expectancy and is scaled by the visit expectancy η_t. The natural-history component TV is evaluated at calendar time. The observed score is their sum plus a correlated error. A blinded-discontinuation block advances calendar time and belief time while freezing time on drug, so the three components are separable by design.

The template is not estimated. The analysis model is linear in calendar time and in the continuous drug indicator $D_{bc,it}$. It has no parameter for response shape. The template is a fixed nuisance structure that enters every cell of every parameter grid. No diagnostic inside a standard run would reveal a poor choice.

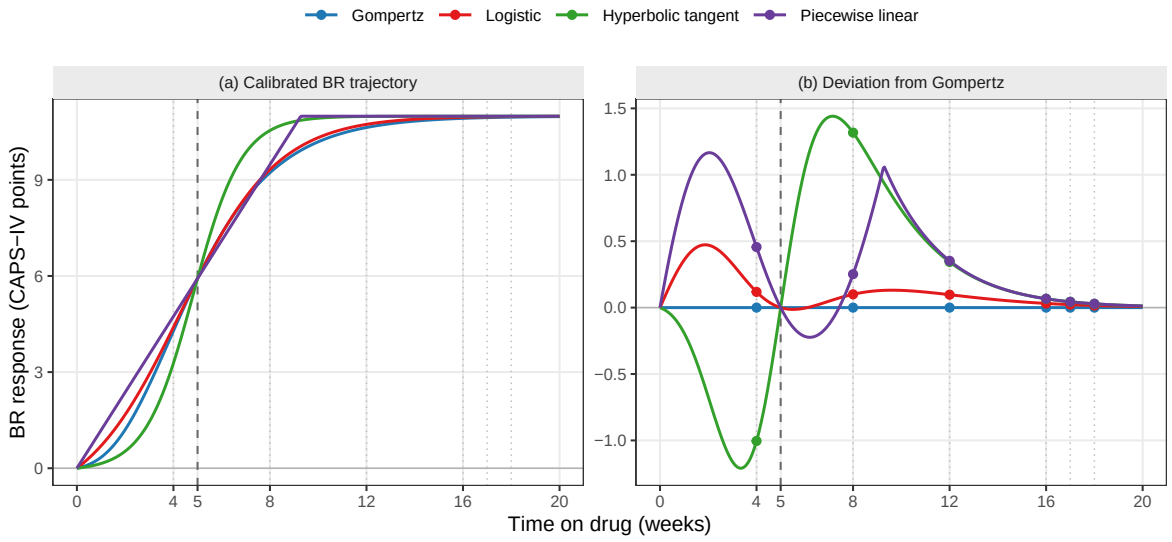


Figure 1: The four trajectory families at the calibration used throughout this paper. Panel (a): the drug-response mean curve under each family against cumulative time on drug, matched at the asymptote and at the week-5 anchor (dashed line). Panel (b): each family's deviation from the Gompertz reference. Dotted vertical lines and plotted points mark the six distinct on-drug durations the two-path OL+BDC design samples (weeks 4, 8, 12, 16, 17, 18). The logistic and Gompertz curves coincide to within 0.12 points at every sampled visit; only the hyperbolic tangent differs by more than one point, at weeks 4 and 8.

Figure 1 shows the four families at the calibration this paper uses. The piecewise-linear curve rises linearly and saturates at week 9.3. The hyperbolic tangent is the most delayed early and the steepest through its inflection. The Gompertz and logistic curves lie between the two and are nearly indistinguishable. All four are pinned to the same asymptote and the same week-5 value, so they cross there, and the room left for them to differ is small. Across the six on-drug durations the design samples, the largest departure of any family from the Gompertz reference is 1.32 points at the first two visits, against a BR residual standard deviation of 8

points. From week 12 onward the largest departure is 0.35 points. The response has saturated by the third on-drug visit, and the blinded-discontinuation block adds visits that are nearly identical in time on drug. The family question at this design is a question about the first eight weeks on drug. Any answer we report is conditional on a visit schedule that samples the discriminating region twice.

1.3 How the framework treats the template

The framework treats the trajectory template as a fixed calibration input[1]. The package ships one response-parameter set as data, the `extracted_rp` object, with one (m, d, r, σ) quadruple per component. Its documentation describes the set as a 'tabula rasa' chosen so that the three components together could plausibly reproduce the authors' prazosin-PTSD randomized trial[8]. The drug-response component has $m_{BR} = 10.99$ CAPS-IV points, $d_{BR} = 5$ and $r_{BR} = 0.42$ per week. The natural-history and placebo-response components share one parameter set, because the documentation declines to assume a difference between them. Published sensitivity analyses sweep the parameters of the Gompertz family. The family itself is held fixed. The sweeps explore the interior of the family and never test its boundary.

The estimand is the biomarker-by-drug interaction, not the treatment effect. The analysis routine returns the coefficient of `bm:Dbc`. The drug main effect is carried as an adjustment and is not reported. Power here means power to detect that the treatment effect varies with the biomarker. The treatment main effect is the subject of the companion manuscript at `analysis/report/04-treatment-main-effect/`.

The template governs the treatment effect: it sets the BR mean curve, which is the D_{bc} channel. Whether the template can reach the interaction depends on how the biomarker moderation is attached to the trajectory. That attachment is the architecture choice, not a property of the template. Under the covariance-moderation architecture of Hendrickson et al. the moderation is a biomarker-response correlation in the covariance matrix, and the template populates only the mean vector. Under the mean-moderation architecture the moderation multiplies the trajectory. We simulate both.

1.4 Why the question matters

First, the family is fixed before any analysis-side question is asked. A simulation that draws every trajectory from a Gompertz curve cannot detect the cost of that choice. Only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the logic of analysis-model misspecification studies[9], applied one level down in the simulation hierarchy. Sigmoidal mixed models fitted directly to longitudinal data show that a logistic-family curve recovers mean levels even under misspecification[10]. The corresponding question for a simulation DGP has not been asked.

Second, the Gompertz family has known identifiability problems in short noisy trajectories. Joint estimation of (m, d, r) is poorly conditioned: the asymptote and rate correlate strongly,

and the displacement is weakly identified when few visits fall in the inflection region.[6] and[7] propose unified-Richards reparameterizations with interpretable and less correlated parameters.[11] formalize the 'sloppiness' diagnostic.[12] fitted eight tumor-growth models to two in vivo systems and ranked them on fit, identifiability and prediction. That comparison has not been made for short N-of-1 trajectories. We do not make it here; see Section 1.6.

Third, the default reflects historical borrowing. The Gompertz form entered the framework from the growth-curve tradition[2,4]. The pharmacology literature defaults to the Hill and sigmoid-Emax family[13,14] and to indirect-response models[15]. The inheritance is not evidence, and the question merits a direct answer.

1.5 What the N-of-1 simulation literature does

Several N-of-1 simulators specify an explicit response form. Percha et al.[16] built a stochastic time-series model with an exponential approach to a treatment-driven target, separate wash-in and wash-out time constants, and released the software. Gärtner et al.[17] simulate aggregated N-of-1 trials with carryover from a causal graph. Liao et al.[18] simulate several lag shapes for a distributed-lag analysis. Wang and Schork[19] derive power under an AR(1) error structure with carryover. The Hendrickson framework[1] introduces the Gompertz template. Each of these fixes one response form. None compares forms. The CONSORT extension for N-of-1 trials[20] and the programmatic statements[21,22] do not address trajectory shape.

In adjacent literatures the comparison is routine. Oncology modelers fit several growth families and compare them on identifiability and prediction[12]. Pharmacometrics has sensitivity studies for sigmoid-Emax and indirect-response models[23,24]. Latent-class growth-trajectory models treat response-curve heterogeneity by mixtures[25]. The present paper applies a four-family comparison to N-of-1 simulation with the Gompertz form as the reference, reported in the ADEMP framework of Morris, White and Crowther[9].

1.6 What this paper contributes

Section 2 characterizes the modified Gompertz family as implemented in pmsimstats. Section 3 presents the three alternatives at matched asymptote and anchor, and shows what the calibration leaves at the sampled visits. Sections 4 and 5 report a Monte Carlo study under common random numbers. Section 6 interprets the result and states its limits.

The sloppiness diagnostic of[11] motivates the identifiability framing above but is not computed here. It is Study 2 of the working protocol (`00-ademp-protocol.md`) and is future work. The full mathematical and historical context is in the companion document `docs/legacy/25-gompertz-model-evaluation.md`.

2 The modified Gompertz function in pmsimstats

The Gompertz curve originates in Gompertz's law of mortality[26] and was first used as a growth curve by[27]. The framework uses the monotone-saturating form. Each latent component (BR, PB, TV) has a mean

$$G(t; m, d, r) = \frac{m \exp(-d \exp(-rt)) - m \exp(-d)}{1 - \exp(-d)}, \tag{1}$$

where m is the asymptote (points), d the displacement (none) and r the rate (per week). The raw curve $m \exp(-d \exp(-rt))$ takes the value $m \exp(-d)$ at $t = 0$. The implementation subtracts that offset and rescales so that $G(0) = 0$ and $G(\infty) = m$. The code names are `maxr`, `disp` and `rate`.

The curve is asymmetric about its inflection at $t^* = \ln(d)/r$. At the reference parameters $t^* = 3.83$ weeks, where the curve has reached 36 percent of its asymptote. It then needs 11.0 more weeks to reach 99 percent. The Gompertz asymmetry is a short acceleration phase followed by a long deceleration tail.

The prazosin reference cell uses the `extracted_rp` set: $m_{BR} = 10.99$ CAPS-IV points, $d_{BR} = 5$, $r_{BR} = 0.42$ per week, and a within-visit residual standard deviation of 8 points. The placebo-response and natural-history components share $m = 6.51$, $d = 5$ and $r = 0.35$. The full set is documented with the `extracted_rp` data object.

3 Alternative response-curve families

Each alternative is parameterized so that its asymptote equals m_{BR} and its value at the anchor $t = 5$ weeks equals the Gompertz value $G(5) = 5.92$ points. The logistic and hyperbolic-tangent rates are fixed at 0.5 per week and the inflection time is solved by `uniroot`. The breakpoint is solved in closed form. The families therefore differ in smoothness and symmetry while sharing the same effect size at the anchor.

3.1 Symmetric logistic

The logistic curve[28] is

$$L(t; m, r, t_0) = \frac{m}{1 - \ell_0} \left[\frac{1}{1 + \exp(-r(t - t_0))} - \ell_0 \right], \qquad \ell_0 = \frac{1}{1 + \exp(r t_0)}, \tag{2}$$

with the offset at $t = 0$ subtracted and the asymptote restored by the rescaling, as in Equation (1). It is symmetric about its inflection at $t_0 = 4.30$ weeks at the calibration. At the calibrated parameters the logistic curve lies above the Gompertz curve on almost the whole range and reaches 50, 90 and 99 percent of the asymptote at 4.7, 8.9 and 13.7 weeks, against 4.7, 9.2 and 14.8 weeks for Gompertz.

Table 1: Drug-response mean (CAPS-IV points) under the Gompertz family at the six on-drug durations the design samples, and each alternative family's deviation from it. The BR residual standard deviation is 8 points.

Time on drug (weeks)	Gompertz	Logistic minus Gompertz	Tanh minus Gompertz	Piecewise minus Go
4	4.28	0.12	-1.00	
8	9.22	0.10	1.32	
12	10.63	0.10	0.34	
16	10.92	0.03	0.07	
17	10.94	0.02	0.04	
18	10.96	0.02	0.03	

3.2 Hyperbolic tangent

The hyperbolic-tangent curve is

$$H(t; m, r, t_0) = \frac{m}{2}\left(1 + \tanh(r(t - t_0))\right), \tag{3}$$

offset and rescaled as above. Because $\frac{1}{2}(1 + \tanh x) = 1/(1 + e^{-2x})$, this is the logistic curve at rate $2r$. The implementation confirms the identity: the largest absolute difference between H at rate 0.5 and L at rate 1.0 with the same t_0 on a grid over 0 to 20 weeks is $1.8e-15$. The family is not a distinct shape. Its role in this study is a logistic curve with twice the calibrated rate, which gives the steepest rise of the three smooth curves. We found no source for a hyperbolic-tangent time-course model in pharmacokinetics and make no such claim.

3.3 Piecewise-linear breakpoint

The piecewise-linear breakpoint curve, formalized in the broken-line regression literature[29], is

$$P(t; m, t_{bp}) = m \min(1, t/t_{bp}), \tag{4}$$

a linear ramp to the breakpoint t_{bp} and constant afterward. The post-breakpoint slope is zero throughout. At the calibration $t_{bp} = 9.28$ weeks. The breakpoint is the only non-differentiable feature across the four families.

3.4 What the calibration leaves at the sampled visits

Table 1 gives the family values at the six on-drug durations the design samples. The logistic curve departs from the Gompertz curve by at most 0.12 points at any sampled visit. The piecewise-linear curve departs by at most 0.46 points. The hyperbolic tangent departs by -1.00 points at week 4 and 1.32 points at week 8. Against a residual standard deviation of 8 points, the design has at most two distinct shapes where the data are observed: the Gompertz

curve, which the logistic and piecewise-linear curves reproduce, and the hyperbolic tangent. This limits what the simulation can find and is the reason for the result in Section 5.

4 Simulation design

Estimand and performance measures. The estimand is the biomarker-by-treatment interaction coefficient $\beta_{bm:D}$ from the linear mixed-effects model fitted by `nlme::lme` with the formula `Sx ~ bm + t + Dbc + bm:Dbc`, a random intercept (`random = ~1|ptID`) and a continuous-time AR(1) residual correlation (`corCAR1(form = ~t|ptID)`). The p-value is the Wald t test from the `nlme` summary, which uses containment degrees of freedom with no small-sample correction. The primary outputs are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$ and Type I error at $\beta_{bm:D} = 0$. Secondary outputs are the mean and empirical standard deviation of $\widehat{\beta}_{bm:D}$, the mean model-based standard error, and bias against the closed-form reference below. Proportions carry binomial MCSEs; means carry the standard error of the replicate mean.

Sign convention. The components are non-negative reductions in symptom severity, so an increase in any component lowers Y. The interaction coefficient is therefore negative, and the moderation parameter is positive.

Reference value. With the analysis on the raw biomarker and $D_{bc,it}$ binary at $t_{1/2} = 0$, the population slope of the on-drug drug response on the biomarker is $-c_{bm}\,\sigma_{BR}/\sigma_{bm} = -0.45 \times 8/15.36 = -0.2344$ points per biomarker unit. This is the conditional mean of BR given B_i under covariance moderation and the shift under constant-scaled mean moderation. Under trajectory-scaled mean moderation the per-visit multiplier is normalized so that its on-drug mean is σ_{BR}, so the same value is the visit-averaged target; the LME's weighting across visits is the only reason it is approximate. We report bias against this value in every alternative cell.

Design and parameters. The trial design is the two-path open-label plus blinded-discontinuation (OL+BDC) design: visits at weeks 4, 8, 12, 16, 17, 18, 19 and 20; open label at the first four visits; blinded discontinuation at week 18 (path A) or week 19 (path B); no crossover. This is not the four-path Hybrid design of the companion manuscripts. $N = 70$ is the total across the two paths, 35 per path. Carryover is absent ($t_{1/2} = 0$) to isolate trajectory shape. The within-component AR(1) correlation is 0.7 and the cross-component correlations are 0.2 at the same visit and 0.1 across visits. The response and biomarker parameters are the prazosin reference sets `extracted_rp` and `extracted_bp`.

Factors. Four families crossed with two architectures at $c_{bm} = 0.45$, and four families at $c_{bm} = 0$. At $c_{bm} = 0$ the two architectures build the same covariance matrix and apply no mean shift, so the null DGP is the same under both. We run it once per family. The driver confirms on 20 seeds that the two architectures return identical fits at the null.

Notation for the moderation parameter. c_{bm} is used as a common label for the moderation strength at a matched numeric value. Strictly, it is the biomarker-response correlation

under covariance moderation and the multiplier β_{bm} under mean moderation. The two are on a common scale because the mean shift is $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker b_i.

Family encoding. The family sets the BR mean curve inside the covariance construction via the `br_family` argument of `buildSigma`. Under mean moderation the interaction is the multiplicative form of the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`, $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} b_i) + \varepsilon_{it}$, so the biomarker-dependent part of the treatment effect follows the response trajectory (`moderation_scaling = 'trajectory'`). The per-visit moderation is normalized so its on-drug mean equals σ_{BR}. Under covariance moderation the interaction is the biomarker-response correlation in the BM-BR block, and the family enters only through the mean curve.

Common random numbers. Every family and both architectures use the same seed for the same replicate index. The seed is set immediately before the multivariate-normal draw, and the covariance matrix has the same dimension in every cell, so the standard-normal matrix Z is identical across families. Family contrasts are therefore paired. For rejection decisions the paired standard error of a difference in proportions is $\sqrt{b + c}/n$, where b and c are the numbers of seeds on which the two families disagree in each direction. For estimates it is the standard error of the mean paired difference. Morris, White and Crowther[9] recommend this design for comparing methods on the same data sets. The same logic applies to comparing DGPs that differ only in a nuisance input.

Replicates. 1,000 seeds per family and architecture at the alternative, and 2,000 seeds per family at the null. Fits are run by `parallel::mclapply` on 8 workers.

The working protocol at `analysis/scripts/gompertz-evaluation/00-ademp-protocol.md` describes a 144-cell grid and two further sub-studies. The executed study is the focused factorial above. Departures from the protocol are recorded in `02-deviations-log.md`. The study is exploratory, and its findings should be confirmed in an independent study.

5 Results

The driver ran 16,000 fits in 7.7 minutes on 8 workers. Convergence was 100.0 percent. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-crn-replicates.rds`; paired contrasts at `07-gompertz-crn-paired.rds`; the cell summary at `07-gompertz-crn-summary.txt`. Every number in this section is computed from those files.

5.1 Power across families under common random numbers

We simulated the eight alternative cells on common seeds and compared each family with Gompertz seed by seed. Table 2 gives the marginal powers and Table 3 the paired contrasts.

Under mean moderation the Gompertz and logistic families returned the same decision in all 1,000 seeds, and the piecewise-linear family differed from Gompertz in 1 seed. The hyperbolic

Table 2: Power at $c_{bm} = 0.45$ by family and architecture, ± binomial MCSE, from 1,000 common seeds per cell. Two-path OL+BDC design, $N = 70$, $t_{1/2} = 0$.

Family	Mean moderation	Covariance moderation
Gompertz	0.782 ± 0.013	0.797 ± 0.013
Logistic	0.782 ± 0.013	0.797 ± 0.013
Hyperbolic tangent	0.772 ± 0.013	0.789 ± 0.013
Piecewise linear	0.781 ± 0.013	0.796 ± 0.013

Table 3: Paired contrasts of each family against Gompertz on the same seeds at $c_{bm} = 0.45$. Power difference ± paired MCSE $\sqrt{b+c}/n$, where b (up) and c (down) count the seeds on which the family rejects and Gompertz does not, and the reverse. Max $|\Delta p|$ is the largest absolute difference in the interaction p-value across seeds. $\Delta\widehat{\beta}_{bm:D}$ is the mean paired difference in the estimate ± its standard error.

Architecture	Family minus Gompertz	Power difference	Discordant seeds (up / down)	Paired z	Max
Covariance	Logistic	0.000 ± 0.000	0 / 0	undefined	
Covariance	Hyperbolic tangent	-0.008 ± 0.003	0 / 8	-2.8	
Covariance	Piecewise linear	-0.001 ± 0.001	0 / 1	-1.0	
Mean	Logistic	0.000 ± 0.000	0 / 0	undefined	
Mean	Hyperbolic tangent	-0.010 ± 0.003	0 / 10	-3.2	
Mean	Piecewise linear	-0.001 ± 0.001	0 / 1	-1.0	

tangent had lower power by 0.010 (paired MCSE 0.003, $z = -3.2$), with 10 seeds rejecting under Gompertz and not under tanh and 0 the reverse. Under covariance moderation the pattern was the same: Gompertz and logistic identical in every seed, piecewise-linear different in 1 seed, and the hyperbolic tangent lower by 0.008 (paired MCSE 0.003, $z = -2.8$). The number of seeds on which any family disagreed with any other was 10 under mean moderation and 8 under covariance moderation. The largest absolute change in the interaction p-value against Gompertz was 0.030 under mean moderation and 0.056 under covariance moderation, both for the hyperbolic tangent.

The family effect is therefore about one percentage point of power, confined to the hyperbolic tangent, and present under both architectures. It matches Section 3.4: the hyperbolic tangent is the only family that differs from Gompertz by more than one point at any sampled visit, and it is the only family that moves the test.

The covariance architecture gave higher power than the mean architecture within every family, by 0.015 to 0.017 (paired MCSE 0.009 to 0.009).

Comparison with the unpaired run. An earlier version of this study (driver `02-faithful-trajectory-sweep.R`, run 2026-09-06) gave each of its 16 cells its own seed stream. Its mean-moderation powers at $c_{bm} = 0.45$ were 0.743, 0.782, 0.760, 0.778 for Gompertz, logistic, hyperbolic tangent and piecewise-linear, a spread of 0.039, and its covariance-moderation powers were 0.774, 0.791, 0.777, 0.804. A difference of two independent proportions near 0.78 at 1,000 replicates has standard error 0.019, so that spread

Table 4: Empirical Type I error at $c_{bm} = 0$ by family, ± binomial MCSE, from 2,000 common seeds per family. The null DGP is the same under both architectures, so each family is run once. The z statistic uses the binomial standard error at the nominal 0.05.

Family	Replicates	Rejection rate	z against 0.05	Mean $\widehat{\beta}_{bm:D}$
Gompertz	2,000	0.034 ± 0.004	-3.3	0.0014 ± 0.0018
Logistic	2,000	0.034 ± 0.004	-3.3	0.0014 ± 0.0018
Hyperbolic tangent	2,000	0.033 ± 0.004	-3.5	0.0014 ± 0.0018
Piecewise linear	2,000	0.034 ± 0.004	-3.3	0.0014 ± 0.0018
Pooled	8,000	0.034 ± 0.002	-6.7	0.0014

was within Monte Carlo noise. The earlier manuscript interpreted it as a family effect with Gompertz the lowest-power family. The paired comparison shows that Gompertz and logistic give identical decisions and that the spread was noise. We retract the earlier interpretation.

5.2 Type I error

We simulated the four null cells on 2,000 common seeds each. Table 4 gives the rejection rates. The rejection rate ranged from 0.033 to 0.034 across families and was 0.034 (MCSE 0.002) pooled over 8,000 replicates, 6.7 standard errors below the nominal 0.05. The families did not differ: at the null no moderation is applied, so the family enters only the BR mean curve, and the families disagreed on the decision in 2 of 2,000 seeds. The test runs below its nominal size, at 68 percent of 0.05. The power figures in Section 5.1 are therefore made at an effective α near 0.034, and would be higher under a test that held its size. The earlier unpaired run gave a pooled rate of 0.027 over its 8,000 null fits on different seeds; the two runs differ by about 2.5 standard errors, which we note without explanation.

The earlier version of this study ran eight null cells, four families by two architectures, without noticing that the two architectures coincide at $c_{bm} = 0$. Its eight rates were four DGPs run twice on different seeds, and the same-DGP pairs differed by up to 0.016. The present design runs each null DGP once on twice as many seeds.

We do not establish the cause of the size deficit. The `nlme` Wald test uses containment degrees of freedom and makes no small-sample correction, so a degrees-of-freedom adjustment of the Kenward-Roger type[30] would make the test more conservative, not less. The deficit must come from elsewhere. Section 5.3 shows that the model-based standard error exceeds the empirical standard deviation of the estimate in every cell, so the Wald statistic is deflated. A plausible cause of that inflation is the mismatch between the working model, a random intercept plus a CAR(1) residual, and the DGP, which has three AR(1) components with a placebo-response variance that scales with expectancy. We have not tested this conjecture. The companion manuscript at `analysis/report/10-interaction-test-calibration/` studies the calibration of this test directly.

Table 5: Mean interaction estimate, bias against the closed-form reference $-c_{bm}\sigma_{BR}/\sigma_{bm} = -0.2344$ points per biomarker unit ($\pm$ MCSE), empirical standard deviation of the estimate, and mean model-based standard error, at $c_{bm} = 0.45$.

Architecture	Family	Mean $\hat{\beta}_{bm:D}$	Bias	Empirical SD	Mean model SE
Covariance	Gompertz	-0.2331	0.0013 $\pm$ 0.0024	0.0761	0.0875
Covariance	Hyperbolic tangent	-0.2330	0.0013 $\pm$ 0.0024	0.0761	0.0880
Covariance	Logistic	-0.2331	0.0013 $\pm$ 0.0024	0.0761	0.0875
Covariance	Piecewise linear	-0.2330	0.0014 $\pm$ 0.0024	0.0761	0.0875
Mean	Gompertz	-0.2355	-0.0012 $\pm$ 0.0025	0.0805	0.0882
Mean	Hyperbolic tangent	-0.2332	0.0012 $\pm$ 0.0025	0.0805	0.0888
Mean	Logistic	-0.2353	-0.0009 $\pm$ 0.0025	0.0805	0.0882
Mean	Piecewise linear	-0.2347	-0.0003 $\pm$ 0.0025	0.0805	0.0882

5.3 Bias and sampling distribution of the interaction estimate

We computed the mean of $\hat{\beta}_{bm:D}$ in every alternative cell and differenced it against the reference value -0.2344. Table 5 gives the result. Bias ranged from -0.0012 to 0.0014 points per biomarker unit, with MCSE 0.0024 to 0.0025; no cell was more than 0.6 standard errors from zero. The mean estimate did not depend on family: the largest paired difference against Gompertz was 0.0023 (Table 3). The empirical standard deviation was 0.080 to 0.081 under mean moderation and 0.076 to 0.076 under covariance moderation; the covariance cells are uniformly tighter, which is why their power is higher. The mean model-based standard error was 0.087 to 0.089, above the empirical standard deviation in every cell; the ratio of the two ranged from 1.09 to 1.16 at the alternative and was 1.10 at the null. A model standard error that overstates the sampling variability deflates the Wald statistic, which is consistent with the size deficit in Section 5.2. At the null the mean estimate was within 0.0014 of zero in every family (Table 4).

6 Discussion

6.1 The supported result

Families that coincide at the sampled visits give the same decisions. The calibrated logistic curve equals the Gompertz curve to within 0.12 points at every visit the design samples, and the two families returned the same decision in every seed under both architectures. The piecewise-linear curve is within 0.46 points and differed in one seed. The hyperbolic tangent, which is the logistic curve at twice the rate, differs by about one point at weeks 4 and 8 and lowers power by about 0.01 under both architectures. That is the whole family effect at this design. It does not depend on the interaction encoding, and its direction is the same under both.

The earlier version of this manuscript reported a 0.039 spread under mean moderation, with Gompertz the lowest-power family, and an exactly inert covariance architecture. Both claims

were wrong. The spread was the noise of independent seeding at 1,000 replicates per cell. The covariance architecture is nearly but not exactly inert (Section 6.2). The method that exposed the error, common random numbers with paired contrasts, is the one that between-DGP comparisons should use.

6.2 Why the covariance architecture is nearly inert

Under covariance moderation the interaction is the biomarker-response correlation in the BM-BR block. The family enters only the BR mean curve. If the mean curve were in the column space of the fixed effects, the interaction test would be exactly invariant to it. It is not. The fixed effects are an intercept, the biomarker, linear time, D_{bc} and their product. The family-specific mean curve is a function of visit and path that these columns do not span. The part they do not span enters the residuals. It therefore enters the REML estimates of the random-intercept variance, the residual variance and the working CAR(1) parameter, and through them the generalized least-squares weights and the test statistic. The effect is small because the unspanned part is small: at most 1.32 points against a residual standard deviation of 8. It is not zero: the families disagreed on the decision in 8 of 1,000 seeds under covariance moderation, and the largest change in the p-value against Gompertz was 0.056.

Under mean moderation the family enters twice: through the BR mean curve, as above, and through the per-visit multiplier on the biomarker-dependent shift. The second channel is where a family effect was expected. It is small for the same reason: the multipliers are normalized to a common on-drug mean, and the families differ in their visit profile by amounts that the Wald test, which averages over visits, mostly removes.

6.3 Design conditionality

The result is a property of a visit schedule with three on-drug visits at or beyond saturation. The design samples on-drug durations of 4, 8, 12, 16, 17 and 18 weeks. The Gompertz curve has reached 39, 84 and 97 percent of its asymptote at the first three. The families differ only at the first two. A design with weekly visits in weeks 1 to 8 would sample the rising phase where the curves differ and could give a different answer. So could a calibration that matched the asymptote and the time to half-maximum but let the early slope differ. We have not run either.

6.4 Conditions that would change the answer

Four conditions, none exercised here, could enlarge the family effect. First, smaller N: the Wald test would integrate over fewer participants, but the visit schedule would be the same, so we expect little change. Second, a different c_{bm}: at a power near 0.5 the test is most sensitive to small shifts in the noncentrality parameter, so the hyperbolic-tangent difference could be larger in absolute terms. We have no cell at another c_{bm} and do not estimate the size. Third, carryover: with $t_{1/2} > 0$ the off-drug mean decays from the last on-drug value, which

differs by family at week 18 by at most 0.03 points, so we expect little change at this design. Fourth, non-monotone shapes: a family that peaks and declines within the trial window would produce a different on-minus-off contrast and would move the test. The four families here are all monotone-saturating. A matched-seed run at $t_{1/2} = 0.5$ weeks and at $N = 35$ would cost about 15 minutes and would answer the first and third conditions directly; we have not run it.

6.5 The prazosin reference cell

For the prazosin-PTSD application that motivates the framework, the present result implies that sensitivity to the Gompertz assumption is small relative to sensitivity to the carryover specification (the subject of `analysis/report/02-carryover-sensitivity/`) and to the interaction encoding (the subject of `analysis/report/01-dgp-mean-moderation-vs-mvn/`). At the reference parameters and this design, trajectory family changes power by about 0.01 at most. Sensitivity-analysis effort should go to the carryover and encoding axes.

7 Conclusions

First, at the two-path OL+BDC design with $N = 70$ and the prazosin reference parameters, the biomarker-by-treatment interaction test is insensitive to the response-curve family to within one percentage point of power. Families that coincide at the sampled visits give identical decisions. The hyperbolic tangent, which is the logistic curve at twice the calibrated rate, lowers power by about 0.01 under both interaction encodings. The modified Gompertz default can be retained.

Second, the covariance-moderation architecture is nearly but not exactly inert to the family. The family-specific mean curve is not spanned by the fixed effects, so it reaches the REML variance components and the test statistic. At this design the effect is a handful of discordant decisions in 1,000.

Third, the test runs below its nominal size at this design, at 0.034 against 0.05. The cause is not the degrees-of-freedom rule, which makes no small-sample correction. The model-based standard error exceeds the empirical standard deviation of the estimate; we conjecture a working-model mismatch as the cause and have not tested it.

Fourth, comparisons between data-generating processes that differ in a nuisance input must be paired by common random numbers. An unpaired comparison at 1,000 replicates per cell has a standard error of about 0.019 on each contrast and produced a spurious 0.039 family effect in the earlier version of this study.

Fifth, the result is conditional on a visit schedule that samples the rising phase of the response at two visits. A design with denser early visits, or a calibration that leaves the early slope free, is the natural follow-up.

Glossary

Terms are listed in the order in which the paper introduces them. Labels in backticks are the identifiers used in the repository's code, data files and table headings.

- **Trajectory template** (`br_family`): the parametric curve that maps elapsed time onto the expected symptom reduction of a latent response component.
- **Modified Gompertz curve** ($G(t; m, d, r)$, `gompertz`): Equation (1), with asymptote m, displacement d and rate r; the framework's default.
- **Mean-moderation architecture** (`mean_moderation`, "Mean" in tables): the biomarker scales the treatment effect in the conditional mean.
- **Covariance-moderation architecture** (`mvn`, "Covariance" in tables): the biomarker-response correlation is present on drug and absent off drug; the interaction is in the covariance matrix.
- **Moderation strength** (c_{bm}, `c.bm`): under covariance moderation, the on-drug correlation between the biomarker and the drug response; under mean moderation, the multiplier β_{bm} on the standardized biomarker; the two are on a common scale.
- **Anchor**: the week-5 value of the Gompertz curve to which every alternative family is matched.
- **Trajectory-scaled moderation** (`moderation_scaling = 'trajectory'`): under mean moderation, the per-visit shift follows the drug-response mean curve, normalized so its on-drug mean is σ_{BR}.
- **Two-path OL+BDC design**: open-label titration at weeks 4 to 16, then blinded discontinuation at week 18 or week 19; two randomization paths; no crossover.
- **Interaction coefficient** ($\beta_{bm:D}$, `bm:Dbc`): the biomarker-by-treatment interaction in the analysis model; the estimand.
- **Continuous drug indicator** ($D_{bc,it}$, `Dbc`): 1 on drug, $e^{-\lambda t_{sd}}$ off drug after exposure, 0 before exposure; binary here because $t_{1/2} = 0$.
- **Reference value**: the closed-form interaction slope $-c_{bm}\sigma_{BR}/\sigma_{bm}$ against which bias is reported.
- **Common random numbers**: the same seed for the same replicate index in every family and architecture, so that contrasts are paired.
- **Discordant seeds**: seeds on which two families reach different rejection decisions; the paired MCSE of a power difference is $\sqrt{b + c}/n$ with b and c the discordant counts in each direction.

Abbreviations

ADEMP: aims, data-generating mechanisms, estimands, methods, performance measures; AR(1): first-order autoregressive correlation; BDC: blinded discontinuation; BR: drug-response component; CAPS-IV: Clinician-Administered PTSD Scale; CAR(1): continuous-time first-order autoregressive correlation; DGP: data-generating process; LME:

linear mixed-effects model; MCSE: Monte Carlo standard error; OL: open label; OL+BDC: open label followed by blinded discontinuation; PB: placebo-response component; PTSD: post-traumatic stress disorder; REML: restricted maximum likelihood; TV: natural-history component.

8 Conflict of interest

The authors declare no conflicts of interest. The author list overlaps with that of the framework evaluated here[1].

9 Funding

This work received no specific funding.

10 Data availability statement

The data and code that support the findings of this study are in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Public release of the repository is pending; the commit reported at rendering should be cited until a tagged release is deposited.

11 Reproducibility

This manuscript was rendered with R version 4.6.1 (2026-06-24) on aarch64-apple-darwin25.4.0. The package set is captured in `renv.lock`. The simulation pipeline uses `nlme` (linear mixed-effects analysis with `corCAR1` residual correlation), `parallel` (`mclapply`), and `data.table`; the manuscript uses `rmarkdown`, `bookdown` and `ggplot2`.

The simulation driver is `analysis/scripts/gompertz-evaluation/03-common-random-numbers-sweep.R`, run from the repository root with `Rscript`. The stored run was generated 2026-10-28 18:03 PDT in 7.7 minutes on 8 workers (16,160 fits), on aarch64-apple-darwin25.4.0 with BLAS libopenblasp-r0.3.34.dylib. It sets the seed to `810001 + i - 1` for alternative replicate i and `820001 + i - 1` for null replicate i, immediately before each multivariate-normal draw, in every family and architecture. The family is routed through the `br_family` argument of `buildSigma`, and the mean-moderation shift is trajectory-scaled by the `moderation_scaling = 'trajectory'` argument of `generateData`; both default to the original behavior, so other pmsimstats runs are unchanged. Outputs are `analysis/data/quick-sim/07-gompertz-crn-replicates.rds`, `07-gompertz-crn-paired.rds`, `07-gompertz-crn-meta.rds` and `07-gompertz-crn-summary.txt`.

A run on another platform is reproducible in its seeds but not draw for draw, because the BLAS and LAPACK differ; contrasts that must be paired have to be run on one platform. The earlier unpaired driver and its outputs (`02-faithful-trajectory-sweep.R`, `07-gompertz-faithful-*`) are kept for the comparison in Section 5.1.

The ADEMP working protocol is at `analysis/scripts/gompertz-evaluation/00-ademp-protocol.md`, and departures from it are recorded in `02-deviations-log.md`. The manuscript source, bibliography (`references.bib`) and CSL file are at `analysis/report/07-gompertz-evaluation/`.

12 References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

2. Laird AK. Dynamics of tumor growth. *British Journal of Cancer*. 1964;13(3):490-502. doi:10.1038/bjc.1964.55

3. Norton L. A Gompertzian model of human breast cancer growth. *Cancer Research*. 1988;48(24 Pt 1):7067-7071.

4. Zwietering MH, Jongenburger I, Rombouts FM, Riet K van 't. Modeling of the bacterial growth curve. *Applied and Environmental Microbiology*. 1990;56(6):1875-1881. doi:10.1128/aem.56.6.1875-1881.1990

5. Richards FJ. A flexible growth function for empirical use. *Journal of Experimental Botany*. 1959;10(2):290-301.

6. Tjørve E, Tjørve KMC. A unified approach to the Richards-model family for use in growth analyses: Why we need only two model forms. *Journal of Theoretical Biology*. 2010;267(3):417-425. doi:10.1016/j.jtbi.2010.09.008

7. Tjørve KMC, Tjørve E. The use of Gompertz models in growth analyses, and new Gompertz-model approach: An addition to the Unified-Richards family. *PLOS ONE*. 2017;12(6):e0178691. doi:10.1371/journal.pone.0178691

8. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133

9. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

537 10. Capuano AW, Wilson RS, Leurgans SE, Dawson JD, Bennett DA, Hedeker D. Sigmoidal mixed models for longitudinal data. *Statistical Methods in Medical Research*. 2018;27(3):863-875. doi:10.1177/0962280216645632

538 11. Jagadeesan P, Raman K, Tangirala AK. Sloppiness: Fundamental study, new formalism and its application in model assessment. *PLOS ONE*. 2023;18(3):e0282609. doi:10.1371/journal.pone.0282609

539 12. Benzekry S, Lamont C, Beheshti A, et al. Classical mathematical models for description and prediction of experimental tumor growth. *PLOS Computational Biology*. 2014;10(8):e1003800. doi:10.1371/journal.pcbi.1003800

540 13. Hill AV. The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves. *Journal of Physiology*. 1910;40:iv-vii.

541 14. Goutelle S, Maurin M, Rougier F, et al. The Hill equation: A review of its capabilities in pharmacological modelling. *Fundamental and Clinical Pharmacology*. 2008;22(6):633-648. doi:10.1111/j.1472-8206.2008.00633.x

542 15. Dayneka NL, Garg V, Jusko WJ. Comparison of four basic models of indirect pharmacodynamic responses. *Journal of Pharmacokinetics and Biopharmaceutics*. 1993;21(4):457-478. doi:10.1007/BF01061691

543 16. Percha B, Baskerville EB, Johnson M, Dudley JT, Zimmerman N. Designing robust N-of-1 studies for precision medicine: Simulation study and design recommendations. *Journal of Medical Internet Research*. 2019;21(4):e12641. doi:10.2196/12641

544 17. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. *BMC Medical Research Methodology*. 2023;23(1):191. doi:10.1186/s12874-023-02012-5

545 18. Liao Z, Qian M, Kronish IM, Cheung YK. Analysis of N-of-1 trials using Bayesian distributed lag model with autocorrelated errors. *Statistics in Medicine*. 2023;42(13):2044-2060. doi:10.1002/sim.9676

546 19. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. *Healthcare*. 2019;7(3):84. doi:10.3390/healthcare7030084

547 20. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004

548 21. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7

549 22. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15

550 23. Ebling WF, Matsumoto Y, Levy G. Feasibility of effect-controlled clinical trials of drugs with pharmacodynamic hysteresis using sparse data. *Pharmaceutical Research*. 1996;13(12):1804-1810. doi:10.1023/a:1016072806164

551 24. Triggs EJ, Charles BG, Contin M, et al. Population pharmacokinetics and pharmacodynamics of oral levodopa in parkinsonian patients. *European Journal of Clinical Pharmacology*. 1996;51(1):59-67. doi:10.1007/s002280050161

552 25. Hockenberry MJ, Hooke MC, Rodgers C, et al. Symptom trajectories in children receiving treatment for leukemia: A latent class growth analysis with multitrajectory modeling. *Journal of Pain and Symptom Management*. 2017;54(1):1-8. doi:10.1016/j.jpainsymman.2017.03.002

553 26. Gompertz B. On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. *Philosophical Transactions of the Royal Society of London*. 1825;115:513-583.

554 27. Winsor CP. The Gompertz curve as a growth curve. *Proceedings of the National Academy of Sciences USA*. 1932;18(1):1-8.

555 28. Verhulst PF. Notice sur la loi que la population suit dans son accroissement. *Correspondance Mathématique et Physique*. 1838;10:113-121.

556 29. Muggeo VMR. Estimating regression models with unknown break-points. *Statistics in Medicine*. 2003;22(19):3055-3071. doi:10.1002/sim.1545

557 30. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558